Jack



Category: **Automotive** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Jack hydraulic; Wheel jack

What it is used for

A jack lifts a vehicle or other approved load for limited access during inspection, wheel changes, or maintenance. Correct load rating, stable positioning, and suitable support stands are essential to prevent the load from falling.

Main parts

- Lifting saddle or pad contacts the approved lifting point.
- Hydraulic cylinder raises the load on hydraulic jack models.
- Pump handle operates the hydraulic lifting mechanism.
- Release valve lowers the load in a controlled manner.
- Base supports the jack on the ground.
- Frame and lifting arm transmit force to the load.
- Wheels or rollers allow positioning on applicable models.
- Load rating label specifies the jack's permitted capacity.

Before you start

1. Read the jack and vehicle manuals and identify the correct lifting points.
2. Check the jack's load rating against the vehicle or load requirements.
3. Inspect the frame, saddle, handle, hydraulic seals, wheels, and release valve for damage or leaks.
4. Park the vehicle on firm, level ground and switch off the engine or drive system.
5. Apply the parking brake and use wheel chocks on wheels remaining on the ground.
6. Keep people clear of the vehicle and prepare correctly rated support stands.

How to use

1. Position the jack on firm, level ground beneath the approved lifting point.
2. Ensure the saddle is centred and makes secure contact with the lifting point.
3. Close the release valve according to the jack instructions before pumping.
4. Operate the handle smoothly and raise the load only to the required height.
5. Check that the load remains stable and the jack stays upright during lifting.
6. Place correctly rated support stands at the manufacturer's approved support points.
7. Lower the load slowly onto the support stands and verify stable contact before working underneath.
8. For lowering, remove people and tools from the hazard area and raise the load slightly to unload the stands.
9. Remove the stands and lower the load slowly using the release valve as instructed.

10. Remove the jack, check for leaks or damage, and store it safely.

⚠ Safety

- Never work under a vehicle supported only by a jack; use correctly rated support stands.
- Never exceed the jack's rated capacity or use it on soft, sloping, or unstable ground.
- Do not place any part of the body under a load while it is being raised or lowered.
- Use only the manufacturer's approved lifting points and keep the saddle correctly positioned.
- Do not use a jack with hydraulic leaks, damaged parts, or an unreadable rating label.
- Keep bystanders clear and never move a vehicle while it is supported by a jack.

Care & storage

- Wipe the jack clean after use and keep the lifting saddle free of debris.
- Inspect hydraulic seals, hoses where fitted, wheels, frame, and release valve regularly.
- Maintain hydraulic fluid only according to the manufacturer's instructions and specified fluid type.
- Store the jack in a dry place and arrange servicing for leaks or faulty lowering.

Common mistakes to avoid

- Lifting from an unapproved vehicle point or on unstable ground.
- Exceeding the jack's load rating.
- Working beneath a vehicle without support stands.
- Opening the release valve too quickly and lowering the load suddenly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.